Computational Complexity (CPS-MIRI)
SAT assignment
Report

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CPS git repository

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Introduction

This is the report for the SAT model to solve the Street Directionality Problem (SDP). The idea is to encode the problem with DIMACS formulas and solve it with the `kissat` SAT solver. The code is in my public GitHub repository.

Variables

Decision Variables

- `E_i,j` (*Boolean*): This variable indicates whether the directed street from crossing i to crossing j is open. A value of 1 means the lane is active, while 0 means it is closed.

Conversion Variables

- `C_m` (*Boolean*): For each original two-way street m, this variable is true when exactly one of its two directions is closed. These are the variables used in the objective function.

Reachability Variables

- `R_s,i,w` (*Boolean*): This variable represents whether crossing i is reachable from source s within accumulated travel time w. These variables are used to encode the QoS constraints.

Cardinality Variables

- `S_m,r` (*Boolean*): Auxiliary variables used by the sequential counter that enforces the minimum number of conversions.

Constraints

Street Integrity

The solver must preserve the original graph structure.

- Non-existent streets are forced to 0.
- Original one-way streets are kept fixed.
- Original two-way streets must keep at least one direction open.

Conversion Encoding

For each original two-way street, the model links the direction variables with the conversion variable:

$$C_m \leftrightarrow \neg(E_{u,v} \wedge E_{v,u}) \tag{1}$$

This means that a conversion is counted exactly when one direction remains open and the other is closed.

Reachability Propagation

The reachability variables are initialized with the source crossing and then propagated through active streets. If a node is reachable with some travel time, then it can also reach its neighbors through an open street while respecting the accumulated cost.

Travel Time Limit

For every source-destination pair (s, d), the final travel time must not exceed the allowed degradation limit:

$$R_{s,d,L_{s,d}} = 1 \tag{2}$$

where $L_{s,d}$ is the maximum allowed travel time computed from the original shortest path and the percentage P.

Cardinality Constraint

The model uses a sequential counter to enforce a minimum number of converted streets. The linear search starts from the maximum possible number of conversions and decreases until the formula becomes satisfiable.

Objective Calculation

The objective is to maximize the number of converted two-way streets. If N is the total number of original two-way streets, then the objective can be written as:

$$\text{maximize} \sum_{m=0}^{N-1} C_m \tag{3}$$

Equivalently, the search checks whether it is possible to convert at least K streets, starting from the largest K and going down until SAT is found.